

tf.compat.v1.profiler.Profiler Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/profiler/Profiler

TensorFlow multi-step profiler.

Function Signatures

```
tf.compat.v1.profiler.Profiler( graph=None, op_log=None )
```

Code Examples

```
tf.compat.v1.profiler.Profiler(  
graph=None, op_log=None  
)
```

```
tf.compat.v1.profiler.Profiler(  
graph=None, op_log=None  
)
```

Typical use case:

Currently we are only allowed to create 1 profiler per process.

```
profiler = Profiler(sess.graph)
```

```
for i in range(total_steps):
```

```
    if i % 10000 == 0:
```

```
        run_meta = tf.compat.v1.RunMetadata()
```

```
        _ = sess.run(...,
```

```
        options=tf.compat.v1.RunOptions(
```

```
        trace_level=tf.RunOptions.FULL_TRACE),
```

```
        run_metadata=run_meta)
```

```
        profiler.add_step(i, run_meta)
```

```
    # Profile the parameters of your model.
```

```
    profiler.profile_name_scope(options=
```

```
    (option_builder.ProfileOptionBuilder
```

```
    .trainable_variables_parameter()))
```

```
    # Or profile the timing of your model operations.
```

```
    opts = option_builder.ProfileOptionBuilder.time_and_memory()
```

```
    profiler.profile_operations(options=opts)
```

```
    # Or you can generate a timeline:
```

```
    opts = (option_builder.ProfileOptionBuilder(
```

```
    option_builder.ProfileOptionBuilder.time_and_memory())
```

```
    .with_step(i)
```

```
    .with_timeline_output(filename).build())
```

```
    profiler.profile_graph(options=opts)
```

```
    else:
```

```
        _ = sess.run(...)
```

```
    # Auto detect problems and generate advice.
```

```
    profiler.advise()
```

Typical use case:

```
# Currently we are only allowed to create 1 profiler per process.
profiler = Profiler(sess.graph)
for i in range(total_steps):
    if i % 10000 == 0:
        run_meta = tf.compat.v1.RunMetadata()
        _ = sess.run(...,
            options=tf.compat.v1.RunOptions(
                trace_level=tf.RunOptions.FULL_TRACE),
            run_metadata=run_meta)
        profiler.add_step(i, run_meta)
    # Profile the parameters of your model.
    profiler.profile_name_scope(options=
        (option_builder.ProfileOptionBuilder
            .trainable_variables_parameter()))
    # Or profile the timing of your model operations.
    opts = option_builder.ProfileOptionBuilder.time_and_memory()
    profiler.profile_operations(options=opts)
    # Or you can generate a timeline:
    opts = (option_builder.ProfileOptionBuilder(
        option_builder.ProfileOptionBuilder.time_and_memory())
        .with_step(i)
        .with_timeline_output(filename).build())
    profiler.profile_graph(options=opts)
else:
    _ = sess.run(...)
# Auto detect problems and generate advice.
profiler.advise()
```

```
add_step(
    step, run_meta
)
```

```
add_step(
    step, run_meta
)
```

```
advise(  
options  
)
```

```
advise(  
options  
)
```

Documentation

TensorFlow multi-step profiler.

Methods

`## add_step`

[View source](#)

Add statistics of a step.

advise

[View source](#)

Automatically detect problems and generate reports.

profile_graph

[View source](#)

Profile the statistics of graph nodes, organized by dataflow graph.

profile_name_scope

[View source](#)

Profile the statistics of graph nodes, organized by name scope.

profile_operations

[View source](#)

Profile the statistics of the Operation types (e.g.

MatMul, Conv2D).

profile_python

[View source](#)

Profile the statistics of the Python codes.

By default, it shows the call stack from root. To avoid redundant output, you may use options to filter as below
options['show_name_regexes'] = ['.my_code.py.']

serialize_to_string

[View source](#)

Serialize the ProfileProto to a binary string.

Users can write it to file for offline analysis by tfprof commandline or graphical interface.

